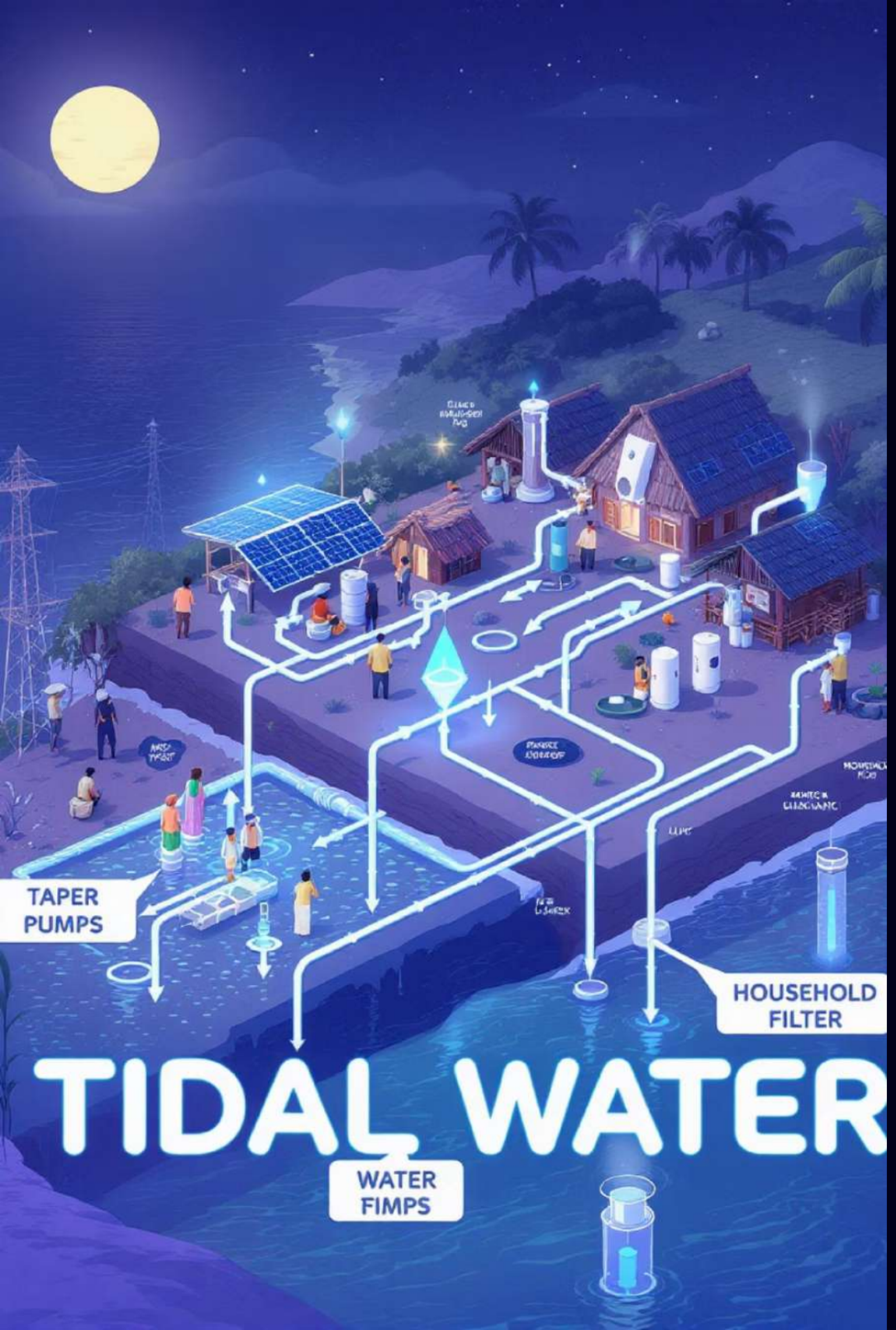
Smart Tidal Water System (STWS)

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 **by MD.Shahriar Shishir Sami**





The Smart Tidal Water System: A Climate-Resilient Solution for Bangladesh

The Smart Tidal Water System (STWS) offers a comprehensive, climate-resilient drinking water solution specifically designed for Bangladesh's coastal regions. It addresses critical challenges such as salinity intrusion, poor infrastructure, and economic constraints, ensuring sustainable and affordable access to safe water.

The Coastal Water Crisis: A Dire Need for Innovation

Coastal Bangladesh is grappling with a severe drinking water crisis. Frequent tidal surges, escalating salinity levels, and the relentless impact of climate change have made safe, potable water an increasingly scarce commodity. Regions like Khulna, Satkhira, Patuakhali, and Bagerhat are particularly vulnerable, with existing infrastructure proving inadequate to meet the growing demand.

Traditional water solutions often fall short due to high electricity consumption, complex maintenance requirements, or prohibitive costs for low-income households. There is an urgent need for a low-cost, locally powered, and climate-resilient water system that can guarantee access for every family, even in flood-isolated or off-grid villages.



Introducing the Smart Tidal Water System (STWS)

The Smart Tidal Water System (STWS) is an innovative three-tier, climate-smart solution designed to address Bangladesh's drinking water crisis. It ingeniously combines natural energy sources–tides and sunlight–with active community participation and cutting-edge water purification technologies.

STWS is built on principles of simplicity and sustainability, making it easy to operate using readily available local resources. It's capable of transforming even highly brackish and saline water into safe, clean drinking water, providing a lifeline to communities most affected by water scarcity.





Tier 1: Tidal-Powered Water Pumps (TPWPs)

Harnessing Tidal Power

TPWPs utilize natural tidal movements in estuaries and canals to lift water without electricity, offering a sustainable, cost-free energy solution.

Local Materials & Resilience

Constructed from bamboo, jute ropes, and reused drums, these pumps are designed for off-grid, cyclone-prone, and power-scarce environments.

Community-Led Operations

Water User Committees (WUCs) manage daily operation and upkeep, supported by Local Government Institutions (LGIs) providing land and initial facilitation.



Tier 2: Solar Village Water Pods (SVWPs) with Desalination

The second tier focuses on purifying and desalinating water drawn by the TPWPs, ensuring it's safe for consumption.

1

Pre-treatment Filtration

Water first passes through Pond Sand Filters (PSF) or Slow Sand Filters (SSF) to remove sediments and biological contaminants.

2

Desalination Processes

Following pre-treatment, water undergoes desalination via solar thermal evaporation or low-energy Capacitive Deionization (CDI).

3

Final Purification

Solar-powered UV disinfection and chlorine dosing eliminate remaining pathogens, ensuring water safety.





Solar Village Water Pods (SVWPs) in breif

Solar-Powered Treatment

SVWPs, strategically located at community hubs like schools, markets, and cyclone shelters, leverage solar energy for advanced water treatment. Water undergoes purification through Pond Sand Filters or Slow Sand Filters.

Advanced Desalination

Desalination is achieved via Solar Thermal Evaporation and Capacitive Deionization (CDI). Final disinfection is ensured through solar-powered UV sterilization and chlorine treatment, guaranteeing safe drinking water.

Smart Monitoring & Engagement

IoT dashboards provide real-time water quality monitoring, ensuring transparency and efficiency. School STEM clubs actively participate in quality checks, fostering local ownership and scientific literacy among students.

IoT Monitoring and Community Engagement



SVWPs are strategically located in accessible community spaces like schools, cyclone shelters, and marketplaces to maximize reach. Water User Committees (WUCs) oversee their management.

Local school clubs actively participate by monitoring water quality using IoT-enabled dashboards, which send real-time alerts to engineers and community leaders. This facilitates timely maintenance, enhances transparency, and fosters local ownership and technological engagement.

Tier 3: Smart Matir Ghora 2.0 (Household Filters)

The Smart Matir Ghora 2.0 ensures safe water access even in the most remote and flood-isolated homes. This household-level solution integrates traditional craftsmanship with modern technology.



Locally Crafted Filters

These filters are crafted by women’s pottery cooperatives using traditional clay designs, promoting cultural acceptance and easy adoption within communities.



Solar UV Sterilization

Each clay filter is paired with a solar UV cap that sterilizes water using sunlight, providing an effective and low-cost method to eliminate remaining pathogens.



Affordability & Empowerment

Priced affordably and subsidized by NGOs, this tier economically empowers women potters while ensuring last-mile water safety for all.

Smart Matir Ghora 2.0 core concept:

The Smart Matir Ghora 2.0 is a revolutionary three-layered clay filter, designed using traditional local pottery techniques and enhanced with modern technology. This innovative filter is powered by solar energy and features a membrane-based Capacitive Deionization (CDI) unit integrated into its middle layer.

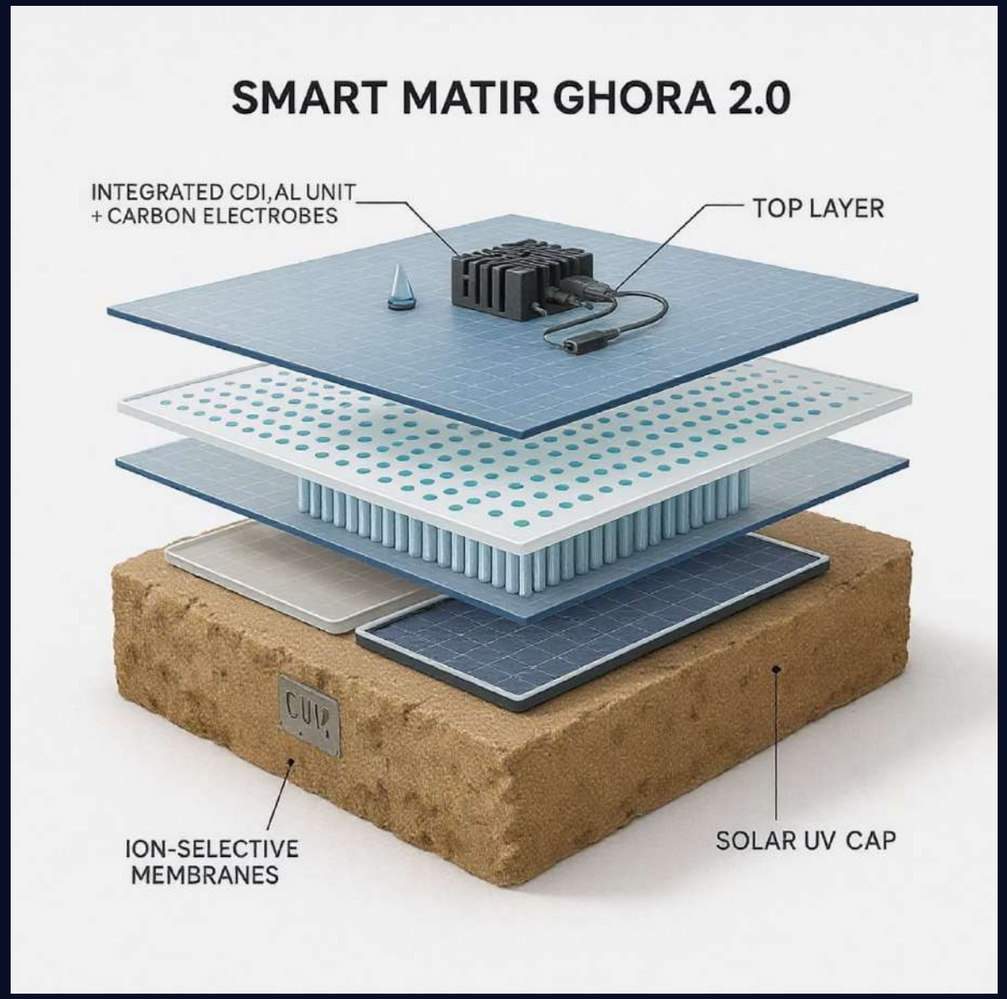
How it Works:

Top Chamber: Brackish or saline water is poured into this chamber.

Middle Chamber: Here, two carbon electrodes, powered by solar energy, utilize ion-selective membranes to efficiently extract salt ions (Na^+ and Cl^-) from the water, without requiring pressure or heat.

Bottom Chamber: Clean, purified freshwater collects in this chamber, ready for consumption.

An integrated solar UV cap further sterilizes the water, ensuring maximum safety and purity.



Innovative Brine Brick Cooperative

Addressing the environmental challenge of desalination byproduct, the system incorporates an innovative Brine Brick Cooperative.



Waste Transformation

Highly saline brine, traditionally a hazard, is mixed with local fly ash, sand, and cement to create eco-friendly construction bricks.

Community-Led Production

Bricks are produced in community-run cooperatives, predominantly led by women's groups and youth entrepreneurs trained in brick molding.

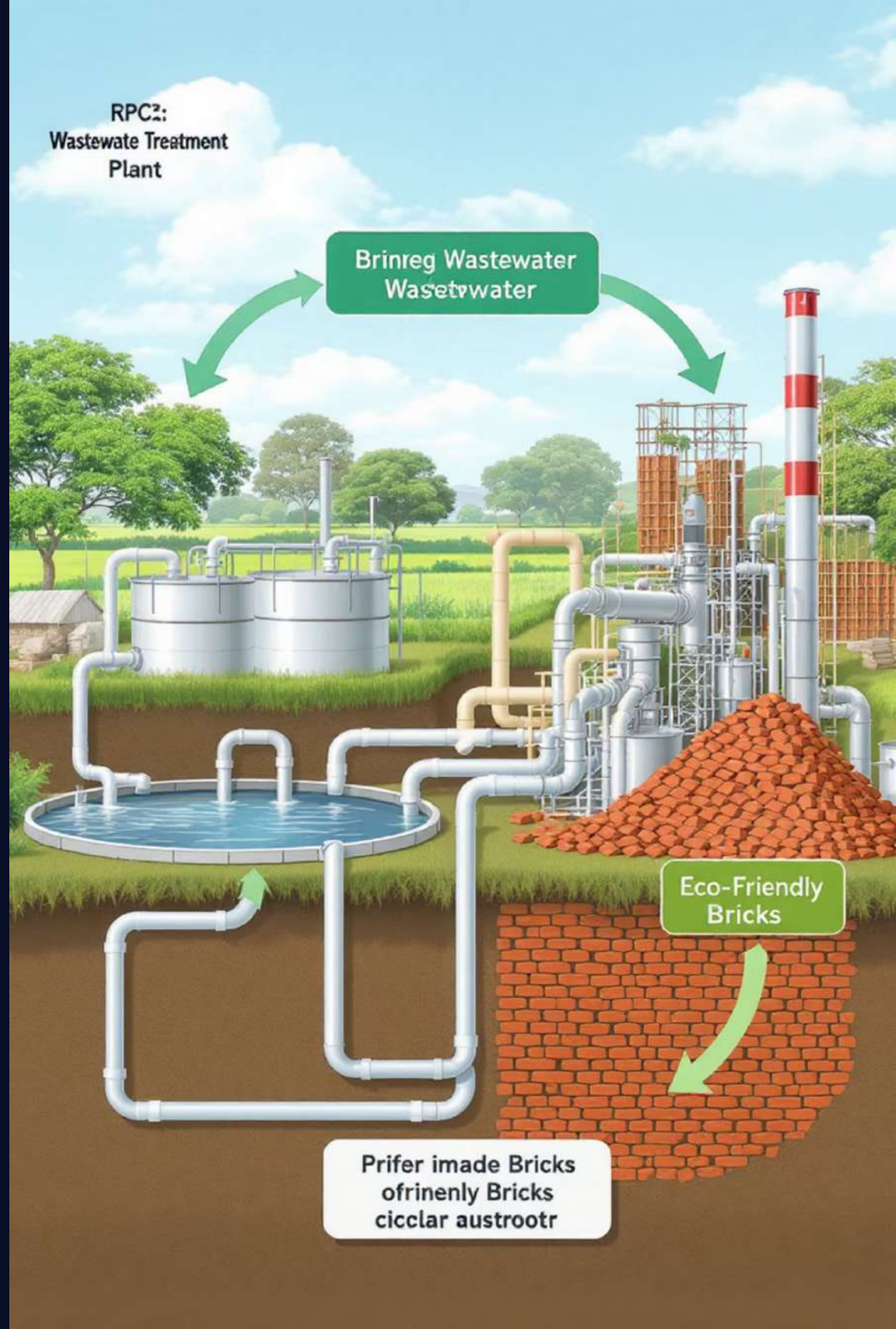
Durable & Sustainable

The resulting durable, salt-resistant bricks are ideal for flood barriers, cyclone shelters, and low-cost housing, transforming waste into economic resources.

Brine Creation and Resourceful Reuse

A unique aspect of the STWS is its responsible management of brine wastewater. After the Capacitive Deionization (CDI) process, a small volume of clean water is used to flush the electrodes, which results in a concentrated salt brine.

This brine, instead of being discarded as waste, is carefully collected in a separate chamber. It is then channeled to the innovative "Brine Brick Cooperative" for valuable reuse, demonstrating a true commitment to circular economy principles and resourcefulness.



The Brine Brick Cooperative: Waste to Wealth



Eco-Friendly Construction

The Brine Brick Cooperative transforms the concentrated brine wastewater into eco-friendly construction bricks. These bricks are made by combining brine with fly ash, cement, and sand, creating a durable and sustainable building material.



Strengthening Infrastructure

The bricks are pressed using manual or low-energy machines, making the production process accessible and sustainable. They are then utilized for vital community infrastructure projects such as flood barriers, cyclone shelters, latrines, and affordable housing, bolstering local resilience.



Job Creation & Empowerment

This innovative not only converts waste into valuable resources but also creates sustainable employment opportunities for local women and youth through cooperative models, fostering economic empowerment and community development.



Local Governance and Maintenance

The system's maintenance and governance heavily rely on active local involvement, ensuring long-term sustainability and ownership.

Water User Committees

Coordinate operations, manage fee collection, and ensure equitable access to water resources.

Youth Repair Squads

Specially trained youth handle technical maintenance of filters, pumps, and digital devices.

Local Government Support

LGIs fund initial setup and integrate the system into regional water and climate resilience plans.

School STEM Clubs

Engaged in monitoring system performance and water quality, fostering environmental stewardship.

Community Governance & Engagement with this system

1

Water User Committees (WUCs)

These committees are at the forefront of managing the day-to-day operations of the STWS, including water distribution and fee collection, ensuring equitable access and sustainability.

2

Youth Repair Squads

Trained youth are equipped with the skills to maintain and repair pumps, filters, solar kits, and IoT devices, ensuring the longevity and continuous functioning of the system.

3

School STEM Clubs

Students actively track water quality, participate in awareness campaigns, and contribute to the scientific understanding of the system, fostering a new generation of environmental stewards.

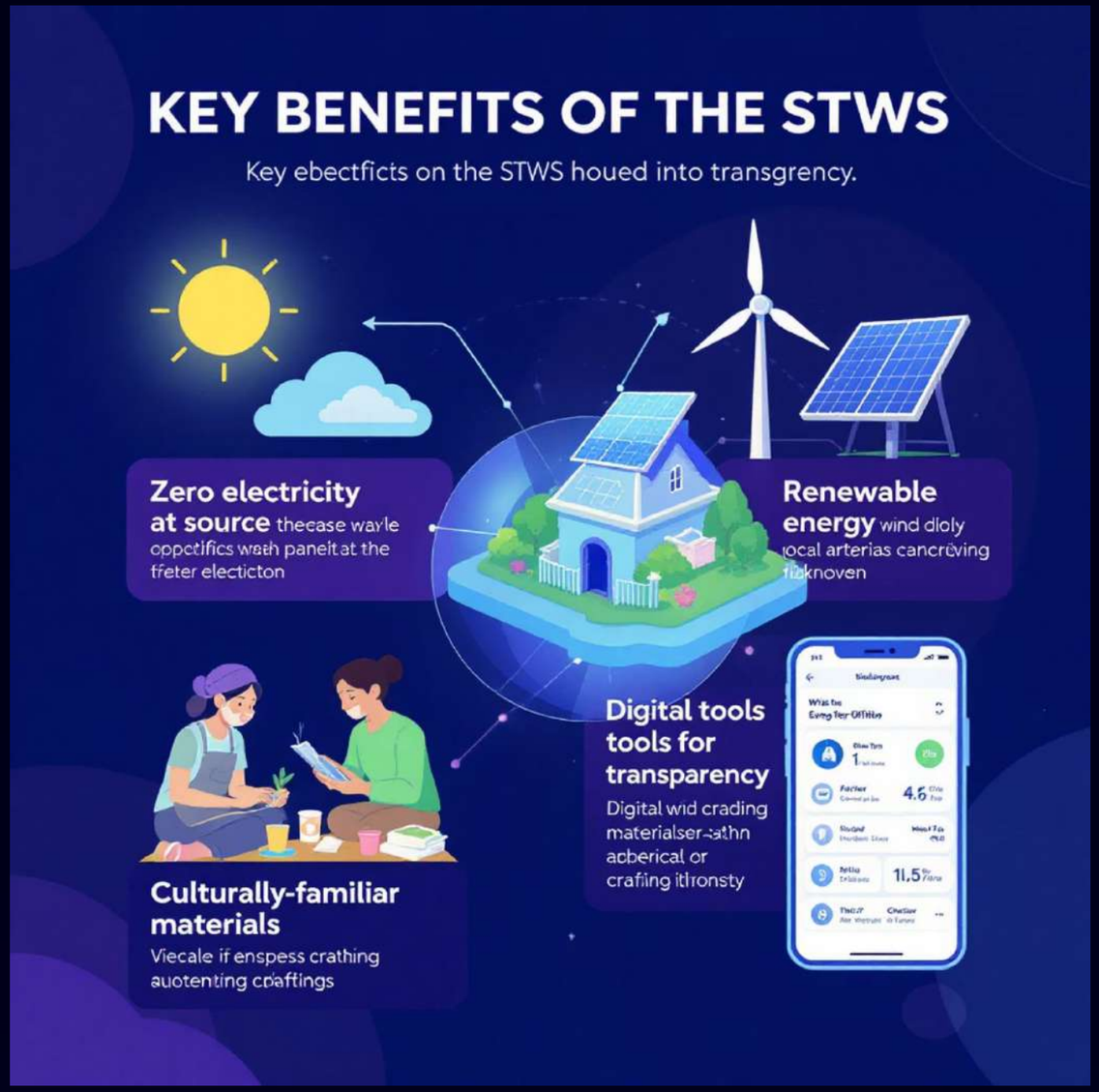
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Local Government Institutions (LGIs)

LGIs provide crucial support by allocating land, securing funding, and integrating the STWS into broader policy frameworks, ensuring its widespread adoption and impact.



Benefits of the STWS Architecture



Energy Independence

Requires no electricity at the water source, relying on natural tidal power.

Renewable Energy

Harnesses solar energy for water treatment processes, ensuring sustainable operations.

Cultural Acceptance

Utilizes culturally familiar materials and designs, promoting ease of adoption.

Digital Transparency

Integrates digital tools for real-time monitoring, transparency, and early warning.

Why STWS Works: A Holistic Solution

Feature	Benefit
Off-grid Power	Harnesses tidal and solar energy, eliminating reliance on conventional electricity, making it ideal for remote areas.
Home Desalination	The Smart Matir Ghora 2.0 with CDI and UV purification provides safe, desalinated water directly at the household level.
Brine Reuse	Transforms a wastewater byproduct into eco-friendly construction bricks, creating economic value and reducing environmental impact.
Community-Run	Empowers women, youth, and schools through active involvement in management, maintenance, and awareness, fostering local ownership.
Scalable Model	Affordable, adaptable, and replicable, allowing for widespread implementation across various coastal regions and beyond.

> The Smart Tidal Water System is more than a technology—it's a movement of resilience, dignity, and empowerment. It works with nature, not against it, and proves that the future of safe water in coastal Bangladesh is not only scalable, it's already possible.



Conclusion: A Holistic and Scalable Model

The STWS positions coastal communities not as passive recipients but as active managers and innovators of their water supply systems. This empowers them to build resilience against climate change, economic limitations, and infrastructural challenges.

> Through sustainable desalination and an innovative brine-to-brick cycle, the STWS presents a holistic, scalable, and socially inclusive model ready for implementation across Bangladesh’s vulnerable delta hotspots.